

Q1: CSP – 5-Queens with Constraint Optimization

Solve the 5-Queens problem:

Tasks:

- **Formulate as CSP:**
 - **Variables = columns**
 - **Domains = rows**
- **Add constraints:**
 - **No same row**
 - **No diagonal attacks**
- **Use:**
 - **Backtracking**
 - **Forward Checking**
 - **MRV**
- **Show:**
 - **Search tree (partial placements)**
 - **Constraint pruning steps**
- **Provide one valid solution.**

Q2: CSP – Resource Allocation with Binary + Ternary Constraints

Variables: {A, B, C, D}

Domains: {1, 2, 3, 4, 5}

Constraints:

- **$A + B = C$**
- **$B < D$**
- **$A \neq D$**
- **$C + D \leq 6$**
- **(A, B, C) must all be different**

Tasks:

- Represent as CSP (include ternary constraint explicitly).
- Convert higher-order constraint into binary constraints if needed.
- Apply Arc Consistency.
- Solve and show reasoning.

Q3: Cryptarithmic

Solve:

SATURN + URANUS + NEPTUNE = PLANETS

Tasks:

- Define all variables and domains.
- Write full column-wise constraints with carries.
- Identify symmetry or pruning opportunities.
- Solve systematically (partial deductions required, full solution not trivial).